

EC9540 - HUMAN COMPUTER INTERACTION

Lab 01

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MediLink: A Community Healthcare Information and Support Platform

Introduction

Healthcare is one of the most important services for every community. However, people living in rural areas of Sri Lanka, often face difficulties in obtaining accurate and timely healthcare information.

Many patients do not know when clinics are available, where medical camps are conducted, whether medicines are available in nearby hospitals during emergencies and get the appointments. Most healthcare information is shared through phone calls, notices, social media or word of mouth, which can be slow and unreliable.

To solve these problems, we propose MediLink.

MediLink is a web-based hospital Medicare management system that allows patients to access healthcare services online. The website provides an easy and user-friendly interface where patients can book appointments, view doctor availability, access medical records, receive prescriptions and make online payments.

The system also helps doctors and hospital staff manage appointments, patient information and medical histories efficiently. The main goal is to improve communication between hospitals and patients while reducing waiting time and paperwork.

The website will be developed with a strong focus on Human Computer Interaction, ensuring that users of all ages can use it easily without confusion.

Description of the Problem.

Healthcare services are essential for ensuring the well-being of individuals. However, many hospitals still face challenges in managing patient services efficiently due to traditional and outdated systems.

- Long waiting times for appointments.
- Difficulty finding available doctors.
- Patients travel to hospitals without knowing whether medicines are available.
- Paper based medical records that can be lost.
- Lack of online appointment booking.
- Limited communication between patients and doctors.
- Time consuming payment processes.
- Difficulty accessing medical history.
- Medical camp information often reaches people too late.
- During emergencies, it is difficult to quickly find blood donors.

- Elderly people often depend on family members or neighbours to obtain healthcare information.
- There is no single platform where people can find local healthcare services.

These problems waste time, increase unnecessary travel, delay medical treatment and reduce access to healthcare services.

Identification of the Target Users

The proposed MediLink - Smart Hospital Medicare Portal is designed to serve different groups of users. Each group interacts with the system differently based on their roles and responsibilities. Understanding these users helps us design an interface that meets their needs and improves the overall user experience.

Primary Users

Primary users are the people who interact with the website regularly and rely on it to perform their daily healthcare related tasks.

1. Patients

Patients are the main users of the system. They use the website to access healthcare services without visiting the hospital for every process.

Connection with the system

- Register and manage their accounts.
- Book, reschedule or cancel appointments.
- View medical records and laboratory reports.
- Receive digital prescriptions.
- Make online payments.
- Receive appointment reminders and notifications.

Dependence on the system

- Quick access to healthcare services.
- Reduced waiting time.
- Easy access to personal medical information.
- Convenient communication with healthcare providers.

2. Doctors

Doctors use the system to provide healthcare services and manage patient information efficiently.

Connection with the system

- View appointment schedules.
- Access patient medical histories.
- Update diagnoses and treatment records.
- Issue electronic prescriptions.
- Review laboratory reports.

Dependence on the system

- Faster access to patient information.
- Better management of appointments.
- Reduced paperwork.
- Improved patient care.

3. Hospital Staff (Receptionists/Nurses)

Hospital staff manage patient related administrative tasks and support hospital operations.

Connection with the system

- Register patients.
- Schedule appointments.
- Update patient information.
- Assist patients with inquiries.
- Manage daily appointment schedules.

Dependence on the system

- Faster patient registration.
- Better appointment management.
- Reduced manual paperwork.
- Improved operational efficiency.

Secondary Users

Secondary users do not use the system continuously but still benefit from its services or support its operations.

1. Parents / Guardians

Parents or guardians manage healthcare services for children or elderly family members.

Connection with the system

- Book appointments on behalf of family members.
- View medical records.
- Receive appointment reminders.
- Make online payments.

Dependence on the system

- Easier management of family healthcare.
- Better communication with hospitals.
- Time-saving access to medical services.

2. Elderly People

Older adults may use the website directly or receive assistance from family members.

Connection with the system

- Book appointments.
- Access prescriptions.
- View medical reports.
- Receive medication reminders.

Dependence on the system

- Easier access to healthcare.
- Reduced hospital visits.
- Better management of chronic illnesses.

Selected Requirement Gathering Method

Selected Method: Questionnaire

Why We Selected a Questionnaire

We selected the questionnaire method because it is suitable for collecting information from many people within a short period of time.

Since our target users include patients, parents, elderly people and healthcare workers, a questionnaire allows us to gather opinions from different groups efficiently.

Questionnaires also make it easier to compare responses because every participant answers the same set of questions.

The collected information will help us understand:

- User needs
- User expectations
- Current healthcare problems
- Features users want
- Difficulties users face
- Preferred interface design

The results will be used to design a user-friendly system.

Advantages of Using a Questionnaire

- Easy to distribute
- Collects responses from many users
- Saves time
- Easy to analyse
- Suitable for different age groups
- Provides consistent data
- Helps identify user needs and expectations before designing the interface.

Prepared Questionnaire

Section A - General Information:

Q1. What is your age group?

- Under 18
- 18 - 30
- 31 - 45

- 46 - 60
- Above 60

Q2. Do you own a smartphone?

- Yes
- No

Q3. How often do you use the Internet?

- Daily
- Weekly
- Rarely
- Never

Section B - Healthcare Experience:

Q4. How often do you visit a hospital?

- Weekly
- Monthly
- Every few months
- Rarely

Q5. How do you currently book hospital appointments?

- Telephone
- Visit hospital
- Online
- Through mobile app

Q6. Have you experienced long waiting times at hospitals?

- Always
- Often
- Sometimes
- Never

Q7. How do you usually receive healthcare information?

- Hospital
- Social Media
- Friends
- Family

- Public Health Midwife
- Other

Q8. How difficult is it to find healthcare information in your area?

- Very Easy
- Easy
- Moderate
- Difficult
- Very Difficult

Section C - System Requirements:

Q9. How satisfied are you with the current appointment booking process?

- Very Satisfied
- Satisfied
- Neutral
- Dissatisfied
- Very Dissatisfied

Q10. Which features would you like to have in the system?

- Clinic schedules
- Medical camp information
- Vaccination reminders
- Health announcements
- Nearby hospitals
- Emergency contacts

Q11. Which language would you prefer?

- English
- Sinhala
- Tamil

Q12. Which device would you prefer?

- Mobile Phone
- Computer
- Both

Section D - Suggestions:

Q13. What healthcare problems do you face most often?

Q14. What additional features would you like to see in this application?

Q15. How easy should the website be to use?

Conclusion

The proposed **MediLink - Smart Hospital Medicare Portal** aims to provide a modern, user friendly and efficient platform for managing hospital services. Through the identification of target users and the collection of user requirements using questionnaires, the project establishes a strong foundation for designing a system that meets the needs of patients, doctors, hospital staff and the wider community. By following Human Computer Interaction principles, the website will focus on usability, accessibility, efficiency and user satisfaction. The information gathered in this phase will guide the design and development of a healthcare system that improves communication, reduces manual processes and enhances the overall healthcare experience.